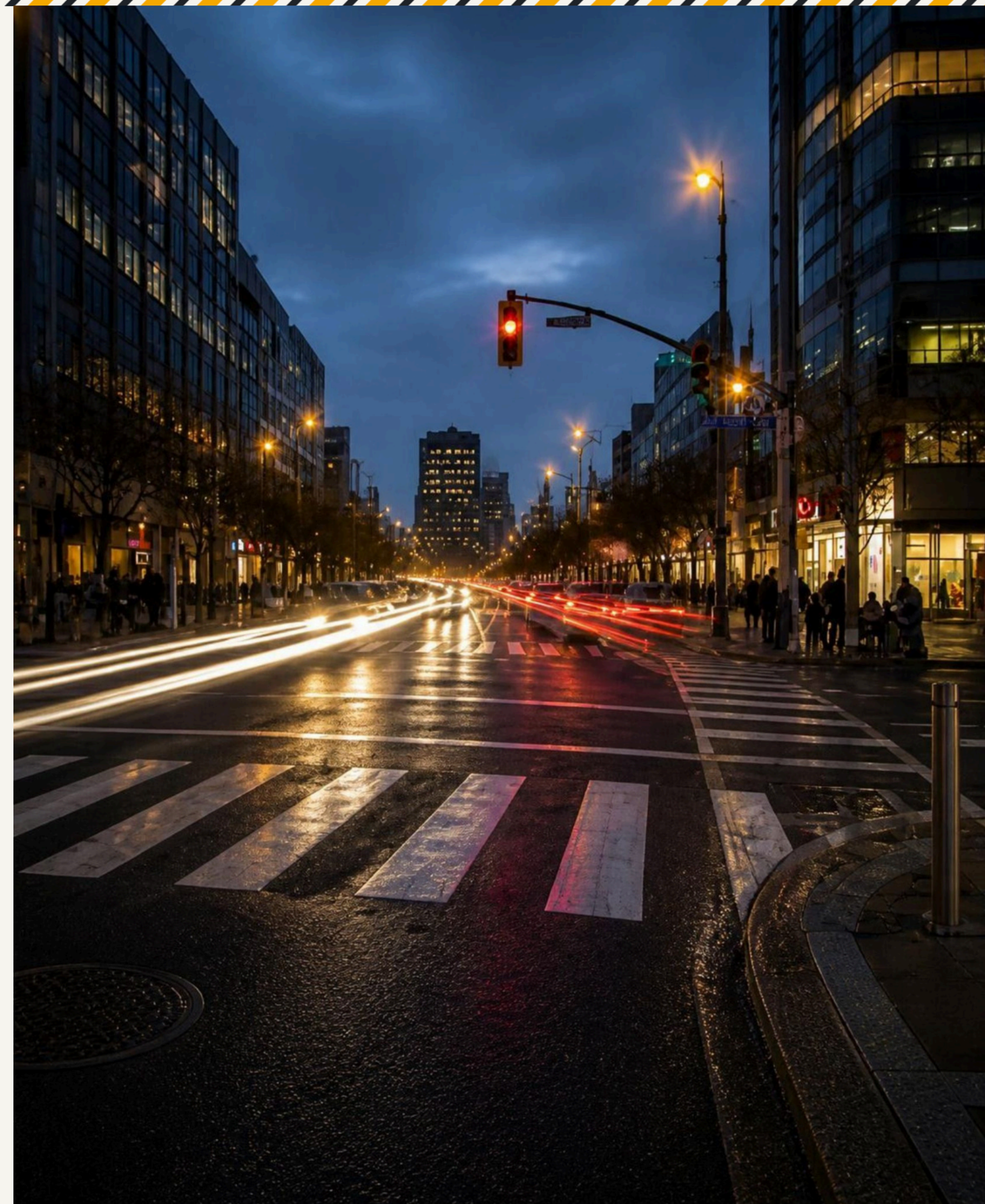
OPEN INNOVATION · CIVIC TECH

RoadPulse

Turning near misses into safer roads

A civic-tech platform that lets everyday observers map traffic hazards **before** crashes happen, turning near misses into explainable risk signals for cities.



Hack The Circle · 2026

THE PROBLEM

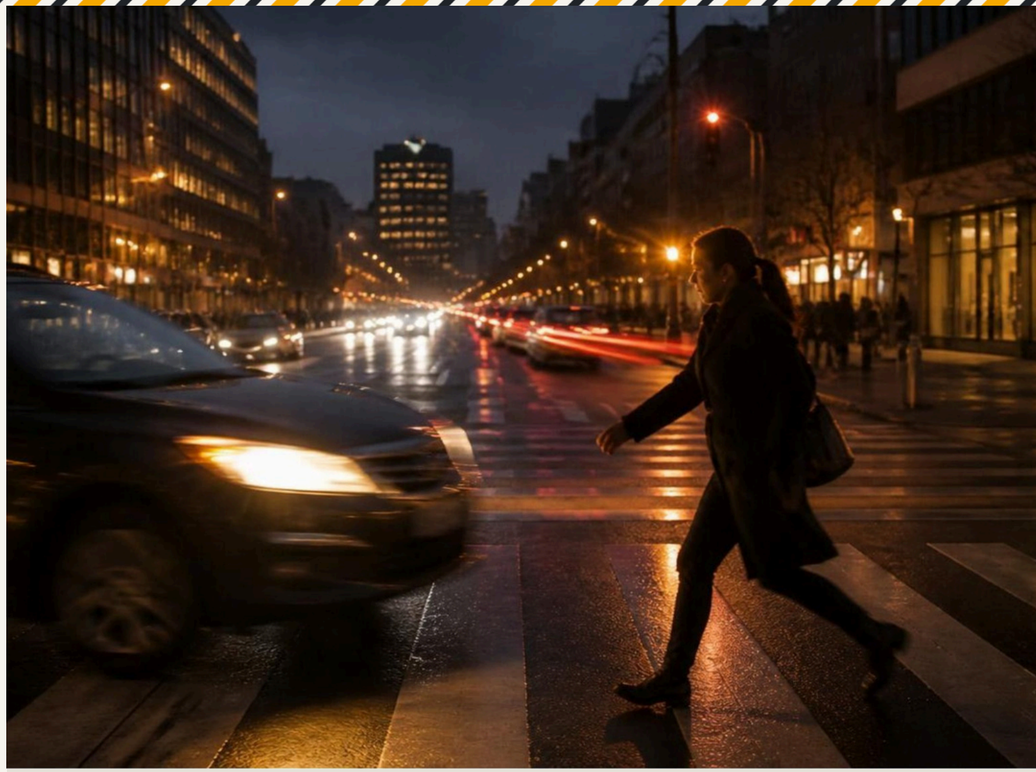
Dangerous moments go unrecorded

No collision, no record. Sudden braking at blind turns, pedestrian near-collisions and unsafe crossings never enter police logs or city safety databases.

Reactive road authorities. Municipal agencies only inspect infrastructure after serious harm occurs, missing recurring near-miss warnings.

Citizen insight stays scattered: eyewitness observations in neighborhood group chats or localized complaints are unstructured, not on actionable geographic maps.

Recurring conflicts repeat unrecorded: vehicle-pedestrian conflicts, poor visibility and unsafe crossings keep happening at the same spots with no shared record.



0 formal logs for near misses today

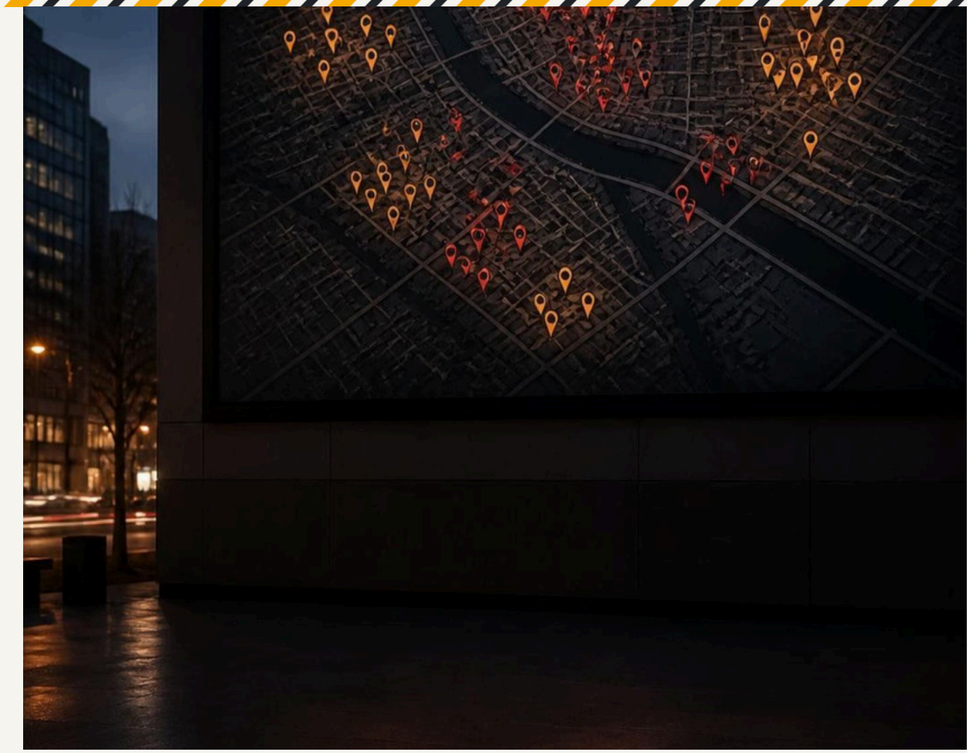
Official crash reporting captures fatalities and property damage, so the preventable near-miss frequency at blind turns and crossings never reaches a safety record.

SOLUTION WORKFLOW

RoadPulse makes observations actionable

An early-warning and decision-support tool, not an accident predictor.

1. Citizens report a near miss in under a minute: device GPS or map pin location, time, hazard type and an optional photo as evidence.
2. Nearby reports cluster by location; repetition, severity, recency, evidence quality and corroboration combine into a transparent, location-level risk score.
3. Safety reviewers verify signals, administrators assign physical site audit tasks, and outcomes are tracked from hazard to resolution, without claiming causation.



4. All of this marks the areas with large near-miss cases as areas needing better infrastructure for prevention of accidents

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CAPABILITIES

Key features built for transparency

Near-miss reporting

Sub-minute submission: map pin or device GPS location, time, hazard type and an optional photo as evidence.

Interactive map

Geospatial heatmaps and multi-factor filters, with unverified reports clearly marked and no personal data shown on the public map.

Explainable risk score

Five weighted factors (repetition, severity, recency, evidence quality, corroboration) in a human-readable score; reviewers see why a location is flagged.

Verification & audits

Reviewer decisions, assigned site audit tasks and outcome monitoring from hazard to resolution, reported without claiming causation.

Area flagging

From the data collected, areas with high near miss cases are flagged for inmprovement in infrastructure/architecture for prevention of accidents.

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SYSTEM ARCHITECTURE

Tech stack: lightweight & proven

Layered architecture of proven open-source tools: transparent, explainable scoring with no custom AI/ML model needed.

CLIENT

React

Tailwind CSS UI

Responsive citizen reporting form and public map portal.

API

Node.js

Express routing

REST endpoints for incident intake and cluster triggers.

DATABASE

PostgreSQL

PostGIS extension

Spatial indexing and coordinate radius clustering.

GEOSPATIAL

Leaflet

OpenStreetMap & Chart.js

Map tile rendering, marker filters, and risk trend charts, area flagging.

STORAGE & AUTH

Supabase

S3 media bucket

Role-based auth for citizens, safety reviewers and administrators, plus photo evidence object storage.

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RISK SCORING

How the explainable risk score works

Score = 0.30R + 0.25V + 0.20T + 0.15E + 0.10C on a 0–100 scale.

Five weighted factors

- Repetition: how often similar reports cluster
- Severity of the reported hazard
- Recency of the reports
- Evidence quality, such as photos and complete logs
- Corroboration by independent reports

Priority bands

- 0–39: routine
- 40–69: elevated
- 70–100: high review priority

Proposed thresholds, not validated standards.

Human in the loop: a score triggers review; it does not prove cause. Reviewers verify the signal before any audit task is assigned.

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RISKS

Risks and mitigations

Risk	Mitigation
False or misleading reports	Verification statuses; unverified reports marked on the map; corroboration built into the score
Limited data	Bands are proposed thresholds; scores flag locations for review rather than drawing conclusions
Privacy	No personal data shown on the public map; role-based access for reviewers and administrators
Misleading scores	Five visible factors; no causation claims; RoadPulse complements official road-safety systems

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TRACK FIT

Selected track: Open Innovation

Cross-sector civic tech
A civic-tech idea that fits no single category: open civic technology, municipal infrastructure planning and public safety data, bridging community input with government response.

Complements official road-safety systems
Not replacing municipal crash databases: an upstream early-detection layer that flags hazards before they show up as crash records, linking citizen observation to municipal audit and repair.

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PROJECT STATUS

Where we are today

Design assets are complete (PRD, scoring model, data schema, user flows); the build sprint is next. No code has shipped yet.

DELIVERED

Design & Architecture

- Comprehensive Product Requirements Document; five-factor explainable risk scoring algorithm.
- PostGIS spatial entity relational schema; end-to-end user flows and audit wireframes.

NEXT SPRINT

Build & Validation

- Project setup, auth integration & database deploy; incident submission forms and map clustering.
- Reviewer dashboard implementation; synthetic test dataset generation & demo run.

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